KAREN SIENA VILLANCIO-WOLTER

10/2026

Curriculum Vitae

Department of Electrical and Computer Engineering, University of Washington
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EDUCATION

University of Washington, Seattle, WA — 2024—Present
PhD Student, Electrical and Computer Engineering, passed Qualifying Exam
Advisor: Dr. Kimberly A. Ingraham

University of Washington, Seattle, WA — 2026
MS, Electrical and Computer Engineering (GPA: 3.95)

University of Florida, Gainesville, FL — 2024
BS, Biomedical Engineering, *summa cum laude* (GPA: 3.91)
Thesis: Optimization of IMU placement for gait analysis at varying walking speeds
Advisor: Dr. Daniel P. Ferris

FUNDING AND AWARDS

Graduate Research Fellowship Program, *National Science Foundation* (NSF GRFP) — 2024—Present
College of Engineering Dean's Fellowship, *University of Washington* — 2024—2025
National Merit Scholarship, *National Merit Scholarship Corporation* — 2020—2024
Benacquisto Scholarship, *Florida Department of Education* — 2020—2024
Bright Futures Scholarship, *Florida Department of Education* — 2020—2024
Undergraduate Research Scholarship, *University of Florida* — 2020—2024

PEER-REVIEWED PUBLICATIONS

R Hybart, **KS Villancio-Wolter**, and DP Ferris (2023). Metabolic cost of walking with electromechanical ankle exoskeletons under proportional myoelectric control on a treadmill and outdoors. PeerJ, 11. https://doi.org/10.7717/peerj.15775

CONFERENCE PRESENTATIONS

KS Villancio-Wolter and KA Ingraham. "Characterizing how exoskeleton control parameters affect human gait features", *Dynamic Walking 2026*, Eatonville, WA (June 2026).

POSTER PRESENTATIONS

A Pfister, **KS Villancio-Wolter,** and KA Ingraham. "Priorities and preferences of adults with spinal cord injury for exoskeleton walking", *University of Washington Center for Research and Education on Accessible Technology and Experiences (CREATE) Research Showcase*, Seattle, WA (May 2026).

KS Villancio-Wolter and KA Ingraham. "Using mixed effects models to explore effects of exoskeleton control parameters on human gait features: A secondary analysis", *University of Washington Electrical and Computer Engineering Research Showcase,* Seattle, WA (March 2026).

KS Villancio-Wolter and KA Ingraham. "Quantifying relationships between exoskeleton control parameters and features of human gait", *Northwest Biomechanics Symposium (NWBS),* Vancouver, BC, Canada (May 2025).

KS Villancio-Wolter, S Arvelo Rojas, N Stafford, DP Ferris, and C Liu. "Evaluating the effects of walking speed and inertial measurement unit placement on joint angles during gait", *J. Crayton Pruitt Family Department of Biomedical Engineering Undergraduate Research Day and University of Florida Spring Undergraduate Research Symposium,* Gainesville, FL (April 2024).

KS Villancio-Wolter, R Hybart, and DP Ferris. "Evaluating statistical correlations between physiological markers and metabolic energy expenditure during walking with a robotic ankle exoskeleton", *University of Florida Spring Undergraduate Research Symposium,* Gainesville, FL (April 2023).

RESEARCH EXPERIENCE

Ingraham Lab, University of Washington 2024—Present
Graduate Researcher
Advisor: Dr. Kimberly A. Ingraham

Project 1: Characterizing how exoskeleton control parameters affect human gait features during walking
- Ran a statistical data analysis using linear mixed effects models to quantify the effects of exoskeleton control parameters on user outcomes (energy cost, ankle kinematics, stride time) to predict optimal walking assistance for multiple objective functions simultaneously
- Presented results at three venues, built an interactive website to explore models, manuscript in progress

Project 2: Establishing powered ankle and hip exoskeleton devices for lab
- Wrote control code (C++) for a robotic ankle exoskeleton; debugged existing software and tested new control algorithms with simulated EMG sensor input signals
- Created an app to visually edit exoskeleton assistance profiles and export settings into control code
- Built and tested a robotic hip exoskeleton; 3D-printed hardware, verified circuit function, designed electrical housing, added soft interface for user comfort, implemented structural improvements

Project 3: Priorities and preferences of adults with spinal cord injury for exoskeleton walking
- Conducted user focus groups with 6 individuals with spinal cord injury to identify preferred exoskeleton interfaces and features
- Presented preliminary findings at UW CREATE Research Showcase

Human Neuromechanics Lab, University of Florida 2021—2024
Undergraduate Researcher
Advisor: Dr. Daniel P. Ferris
Mentors: Dr. Rachel Hybart, Dr. Chang Liu

Project 1: Optimization of IMU placement for gait analysis at varying walking speeds
- Designed a protocol to collect IMU, motion capture, and force plate data from participants (N=4) wearing 13 IMUs across 8 walking conditions
- Wrote MATLAB scripts to process and sync Motive and IMU data
- Built OpenSense and Visual3D pipelines to calculate joint angles from IMU and motion capture data
- Performed statistical parametric mapping (SPM) to identify regions of significant differences between IMU and motion capture joint angle calculations at different speeds and IMU configurations
- Wrote and defended an undergraduate honors thesis on project

Project 2: Metabolic cost of walking with electromechanical ankle exoskeletons under proportional myoelectric control on a treadmill and outdoors
- Collected human subject data (N=12) over 4 training sessions with myoelectric ankle exoskeletons
- Analyzed motion capture data to produce lower body joint angles, moments, and powers
- Edited Python scripts to optimize exoskeleton controller interface
- Led to a peer-reviewed publication

Project 3: Evaluating statistical correlations between physiological markers and metabolic energy expenditure during walking with a robotic ankle exoskeleton
- Wrote Python programs to perform time series analyses on multiple physiological datasets (metabolic cost, heart rate, EMG)
- Calculated correlation coefficients between physiological markers and metabolic energy expenditure to identify whether the markers were accurate indicators of metabolic patterns

OrthoBME Lab, University of Florida 2021
Undergraduate Researcher
Advisor: Dr. Kyle D. Allen
Mentor: Dr. Carlos Cruz

Project 1: Analyzing gait patterns in rodents to study preclinical models of osteoarthritis symptoms
- Performed gait analyses in MATLAB from rodent walking trials for osteoarthritis studies

WORK EXPERIENCE

Manufacturing Engineering Intern, Medtronic Neurovascular Division 2023
- Co-led two improvement projects to reduce Solitaire (Revascularization Stent) rework rates by 10.6%
- Piloted a Gage R&R study to investigate root causes of a prevalent rework code, trained and worked with manufacturing operators to complete study, presented technical results to leadership

Clinical Intern, Hanger Clinic: Prosthetics and Orthotics 2022
- Built and adjusted prostheses using plaster molding, plastic pulling, and carbon fiber lamination
- Shadowed prosthetists and occupational therapists during patient visits for prosthetic limbs and braces

TEACHING AND MENTORSHIP

Instructor, University of Washington Science Explorers 2026
- Planned and led engineering lessons for 4th and 5th grade students in an after-school science program

Program Lead, University of Washington Society of Women Engineers (SWE) 2025—2026
- Mentored 7 undergraduate engineering students and coordinated monthly community events

Research Project Manager, University of Florida Generational Relief in Prosthetics (GRiP) 2022—2024
- Created curriculum for and led technical workshops (PCB Design, KiCad, SolidWorks)
- Ran biweekly meetings with 7 design team captains to monitor project progress and provide feedback
- Held weekly office hours for all organization members (200+ students)

SERVICE AND OUTREACH

Volunteer, University of Washington Go Baby Go (GBG) 2025—Present
- Adapted ride-on cars into pediatric mobility devices for children with disabilities

Volunteer, University of Washington HuskyADAPT 2024—Present
- Led an accessible switch building event for a local Girl Scouts chapter
- Participated in a hackathon to customize accessibility tools for community members

Volunteer, American Society of Biomechanics (ASB) 2022—Present
- Led interactive EMG demos for K-8 students at various outreach events

Volunteer, University of Florida GatorTRAX 2024
- Helped run lessons for an engineering outreach program for elementary school students

Design Team Captain, University of Florida Generational Relief in Prosthetics (GRiP) 2021—2023
- Adapted 70+ toys to send to families of children with disabilities
- Mentored middle school students to design prostheses for stuffed animals on Tinkercad
- Led a design team to create 3D-printed prostheses for 3 canines with limb differences
- Secured $2,000 in project funding from the Donovan J. Welch Fellowship

SKILLS AND CERTIFICATIONS

SolidWorks Certification: Associate - Mechanical Design
Software: Python, MATLAB, C++, JavaScript, HTML, PyTorch, SolidWorks, Cura, Eiger, Motion Capture (OpenSim, Visual3D, Motive), KiCad, MS Office
Instrumentation: Sensors (IMUs, EMG, EEG, ECG, Force Plates, Loadsol, FSRs, Instrumented Treadmills, COSMED), Circuit Building, PCB Design, Machine Shop Tools, 3D-Printing